

Novel Meta Mode-Adaptive Multihead Attention for Multimode Industrial Process Soft Sensing

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Abstract—Soft sensing technologies play a crucial role in industrial production. However, industrial process data typically exhibit complex multimode characteristics, which are often ignored by conventional soft sensing approaches that treat the data as originating from a single mode. To address this limitation, a novel meta mode-adaptive multihead attention (M-MAMHA) method is proposed for multimode soft sensing tasks. First, a MAMHA mechanism is employed to capture dynamic features from each mode while modeling intermode dependencies. The inclusion of adaptive weighting allows the model to assess the importance of each mode automatically, improving flexibility and feature fusion effectiveness. Next, a meta for multimode soft sensor (Meta4MSS) framework based on the Reptile meta-learning algorithm is designed, incorporating a parameter-specific adaptive learning rate. This supports the model in quickly adapting to distributional shifts and diverse modes across tasks, significantly enhancing generalization performance. The proposed method combines the strengths of attention and meta-learning, offering a flexible and robust solution for handling complex multimode industrial data. Experimental validation on two real-world industrial process datasets demonstrates that M-MAMHA achieves superior prediction accuracy and improved robustness compared to state-of-the-art soft sensing models. These results confirm the empirical effectiveness and broad usability of our approach in multimode industrial environments.

Index Terms—Gaussian mixture, meta-learning, multihead self-attention (MHA), multimode industrial process, soft sensor.

I. INTRODUCTION

WITH the rapid development of modern industry, accurately measuring key quality variables has become essential for guaranteeing production process safety and product quality. However, due to the complexity of industrial processes and the limitations of traditional physical sensors, it is often challenging to measure some key quality variables directly [1], [2]. As a result, soft sensing technologies have emerged to predict these variables based on easily detectable process variables, gaining widespread application [3], [4]. Soft sensing modeling methods are primarily divided into two categories: mechanism modeling and data-driven modeling [5]. Traditional mechanism modeling relies entirely on the physical equations of industrial processes. This approach is

challenging to apply to highly complex modern industrial systems. On the other hand, driven by the advancement of storage technologies and big data, data-driven modeling has become more prevalent in modern industrial processes [6], [7], [8]. In data-driven modeling approaches, the artificial neural network can automatically learn nonlinear mappings from data and adapt to dynamic and complex datasets [9], [10]. As artificial intelligence rapidly evolves, simple neural network architectures often lack sufficient expressiveness. This has led to increasing research in applying deep learning to soft sensing modeling [11], [12], [13]. Moreover, industrial processes are typically continuous, and there exists an association among data collected at adjacent sampling times. Thus, considering the dynamic characteristics of process information is crucial for accurately predicting key quality variables [14], [15]. Recently, multihead self-attention (MHA) mechanisms have gained popularity in the field of capturing the dynamic characteristics in soft sensor applications. Su et al. [16] utilized MHA to capture key spatiotemporal feature information, combined with the broad learning system, achieving good performance in chaotic time-series prediction. Chen et al. [17] proposed a spatiotemporal multiscale correlation embedding framework based on self-attention mechanisms that accurately models global and local spatiotemporal coupled correlations under multiscale spatiotemporal relationships.

In industrial processes, multimode often arises due to various factors such as diverse input sources, complex interactions, or human factors that contribute to the complexity and variability of the systems involved [18]. These modes may represent different performance levels, operational efficiencies, or product qualities. Therefore, identifying and coordinating these modes is essential to ensure efficiency, quality, and safety in operations. But using traditional soft sensing models to treat multimode industrial processes as single-mode can lead to inaccuracies, loss of critical information, poor adaptability, and an oversimplified understanding of the process. These limitations significantly affect decision-making and overall process performance [19], [20]. So, many data-driven soft sensing modeling methods have been proposed to address multimode industrial issues [21]. Chai et al. [22] proposed a deep probabilistic transfer regression (DPTR) framework that learns Gaussian-based latent feature representations of multimode data. Then, a probabilistic latent space transfer strategy is employed to effectively extract and transfer mode-specific knowledge [22]. Zhang et al. [23] introduced the information complementary fusion stacked autoencoder (ICF-SAE), which incorporates top-down and bottom-up fusion pathways along with a gating mechanism to regulate information flow, thereby enhancing the modeling of multimode characteristics.

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However, these methods did not consider the presence of multiple probability distributions within multimode data. To address this issue, a Gaussian mixture has been introduced, which enables a more accurate representation of mode-specific distributions. By mapping complex data to a mixture of Gaussian components, models can effectively capture diverse mode characteristics and enhance prediction performance. Zhang et al. [24] aligned conditional and marginal probability distributions within the probabilistic latent space of a proposed Gaussian mixture conditional variational autoencoder to mitigate virtual drift and concept drift in multimode processes. Yao et al. [25] proposed a semi-supervised deep dynamic latent variable model (DMVAER) that employs a two-layer latent structure governed by Gaussian mixture distributions to effectively capture mode-specific information and dynamic process characteristics. Cao et al. [26] applied change point detection to dynamically identify shifts in operating conditions and then used a Gaussian mixture model (GMM) to cluster the data into distinct modes. A dedicated subsoft sensor model is trained for each mode to accurately capture mode-specific relationships. Moreover, Zhang et al. [27] introduced a mixture of multivariate Laplacian distributions into the latent space of a variational autoencoder to enhance robustness against outliers and capture the multimode nature of the data. Although recent studies have utilized mixture-based probabilistic models to capture mode-specific information, existing methods still face several limitations. Most notably, these methods lack mechanisms to adapt to the diverse distributions of different modes, which limits their generalization performance on complex multimode industrial data. Furthermore, in industrial process environments, these methods often ignore the inherent temporal dynamics of data, limiting the ability to model complex industrial process data.

Meta-learning is an innovative learning strategy aimed at developing models that can comprehensively and effectively extract the inherent features of data while quickly adapting to varying data distributions [28]. This ability makes meta-learning inherently well-suited for multimode industrial processes. Reptile is a model-agnostic meta-learning algorithm that employs a simpler update strategy to acquire an initial set of model parameters [29]. Under the Reptile framework, training on data from different modes separately, it uncovers shared features and dependencies among these modes to facilitate parameter updates. As a result, the initial parameters obtained through this approach demonstrate strong generalization capabilities in complex multimode industrial processes.

To address the limitations of existing methods in capturing complex mode-specific information and learning dependencies among dynamic process variables, a new approach called meta mode-adaptive multihead attention (M-MAMHA) is developed for multimode data soft sensing. In the application of M-MAMHA, process data is first categorized based on operating conditions or modes to create meta-training tasks for a meta-learning framework. The trained model is then used for predictive evaluation. The effectiveness of the proposed M-MAMHA has been confirmed via experiments on two industrial datasets: the sulfur recovery unit (SRU) process [3] and the combined cycle power plant (CCPP) [8]. The primary contributions of this study can be outlined as follows.

- 1) For the first time, the meta-learning algorithm Reptile is introduced into multimode soft sensing. By treating each operating mode as an independent meta-task, the proposed approach alleviates the limited generalization of conventional soft sensors and enables rapid adaptation to unseen mode distributions, thereby significantly improving robustness and cross-mode transferability.
- 2) A novel MAMHA model is developed for soft sensing applications. Built upon an MHA-based encoder-decoder architecture, MAMHA effectively captures temporal dependencies in industrial process data. Moreover, Gaussian distribution modeling is incorporated into the feature space to characterize distributional properties and extract mode-specific information, while an adaptive weighting mechanism is introduced to guide the learning process. These designs jointly enhance feature representation and prediction accuracy.
- 3) A meta-learning framework for multimode soft sensing, termed meta for multimode soft sensor (Meta4MSS), is proposed based on the Reptile algorithm. In the meta-training stage, parameter-specific learning rates are assigned to each mode subtask and adaptively updated according to the inner-loop loss. This strategy enables efficient modeling of both mode-specific characteristics and shared patterns across modes, leading to a more generalizable initialization and improved performance across diverse operating conditions.

The remainder of this article is structured as follows. The related research is introduced in Section II. The structure of the proposed model, including the detailed description of each module, is given in Section III. The datasets used in this study, as well as the experimental setup and the corresponding results, are introduced in Section IV. In the end, the conclusion of the article is summarized in Section V.

II. RELATED RESEARCH

A. Gaussian Mixture Model

The GMM serves as a probabilistic model that can be viewed as a combination of K individual Gaussian distributions [30], [31]. Each Gaussian component represents a cluster or mode within the data, making GMM particularly suited for handling data with complex, multimode distributions. The probability distribution and probability density function of GMM can be expressed as follows:

$$p(x|\theta) = \sum_{k=1}^K \pi_k \phi(x|\theta_k) \quad (1)$$

$$\phi(x|\theta_k) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_k|^{\frac{1}{2}}} \exp \left[-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right] \quad (2)$$

where π_k refers to the mixing weight of the k th Gaussian component, satisfying $\pi_k \geq 0$ and $\sum_{k=1}^K \pi_k = 1$. $\phi(x | \theta_k)$ denotes the probability density function of the k th Gaussian component, where $\theta_k = \{\mu_k, \Sigma_k\}$ represents the mean and covariance of the k th Gaussian component, respectively. The density $\phi(x | \theta_k)$ can be expanded as shown in (2), with d indicating the dimensionality of the data.

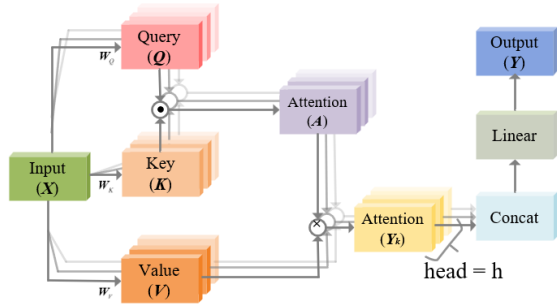


Fig. 1. Flowchart of MHA.

The GMM typically utilizes the expectation–maximization (EM) algorithm for parameter estimation. This algorithm combines maximum likelihood estimation with iterative refinement, allowing it to effectively estimate the underlying data distribution in an unsupervised learning context. Through iterative updates, the algorithm converges to a solution that best fits the data, making GMM a powerful tool for tasks like clustering, density estimation, and anomaly detection.

B. MHA Mechanism

Driven by the Transformer, the self-attention mechanism has become one of the hottest topics in deep learning research in recent years [32]. Due to the ability of the self-attention mechanism to capture correlations within sequential data and assign varying levels of focus to them, it has been widely applied in various fields. MHA further expands this concept by dividing the model into multiple independent heads, each responsible for handling different subspaces of the data and possessing its own weight matrix. The final output of MHA is obtained by merging the outputs of each head. In MHA, the input sequence is transformed into **Query**, **Key**, and **Value** through three sets of linear transformations in (3)–(5). Then, the attention scores between **Query** and **Key** are calculated through (6) and (7), and the weighted output of each head is computed. Finally, the outputs of multiple heads are merged and passed through a linear layer in (8) to obtain the attention-weighted output of MHA. The flowchart of MHA is shown in Fig. 1

$$\mathbf{Query}_k = \mathbf{X} \cdot \mathbf{W}_{Q,k} \quad (3)$$

$$\mathbf{Key}_k = \mathbf{X} \cdot \mathbf{W}_{K,k} \quad (4)$$

$$\mathbf{Value}_k = \mathbf{X} \cdot \mathbf{W}_{V,k} \quad (5)$$

$$\mathbf{Attention}_k = \text{softmax} \left(\frac{\mathbf{Query}_k \cdot \mathbf{Key}_k^T}{\sqrt{d_k}} \right) \quad (6)$$

$$\mathbf{Y}_k = \mathbf{Attention}_k \cdot \mathbf{Value}_k \quad (7)$$

$$\mathbf{Y} = \text{concat}(\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_h) \cdot \mathbf{W}_O \quad (8)$$

where $\mathbf{X}$ and $\mathbf{Y}$ represent the original input sequence and the weighted output sequence, respectively. $\mathbf{W}_{Q,k}$, $\mathbf{W}_{K,k}$, and $\mathbf{W}_{V,k}$ are the weight matrices in linear layers mapping the input features for the k th head. d_k is the vector dimension. h denotes the number of heads, and $\mathbf{W}_O$ is the weight matrix in the linear layer of the output.

This MHA mechanism strengthens the ability of the model to handle complex sequential data, as well as improves its performance and flexibility across various tasks.

C. Reptile

Reptile is a first-order gradient-based meta-learning algorithm designed to initialize the parameters of a neural network in a way that permits the model to swiftly adjust to novel tasks with minimal gradient updates [29]. This enables the model to generalize from a small number of instances in the testing task. An overview of the Reptile algorithm procedure is given below.

- 1) *Initialization*: Begin by initializing the model parameters.
- 2) *Task Preparation*: Randomly sample tasks from the task distribution to construct the meta-training set and meta-testing set.
- 3) *Inner Loop Updates*: Perform multiple gradient updates on the model using the training examples and labels from the sampled task. For each sample, assess the loss, calculate gradients, and modify the model parameters accordingly.
- 4) *Outer Loop Updates*: Once the inner loop updates are completed, update the global model parameters through the difference between the initial parameters and updated parameters.
- 5) *Repetition*: Repeat steps 2)–4) by continuing to sample new tasks and performing the inner and outer loop updates, until the model converges on the task distribution.

III. METHODOLOGY

This section outlines the proposed M-MAMHA method in detail. The overall architecture of M-MAMHA is provided in Fig. 2. By integrating meta-learning with a mode-adaptive attention mechanism, M-MAMHA effectively extracts mode-specific features and models intermode dependencies, thereby improving the generalization capability of the model in complex multimode soft sensing tasks. First, the proposed MAMHA module is introduced in detail, which enables adaptive feature representation for each mode and cross-mode information interaction. Then, the specific implementation steps of the Meta4MSS framework are described, which support task-level adaptation and optimization across diverse mode distributions. Finally, the deployment strategy of the complete M-MAMHA method is discussed, illustrating how the method can be successfully utilized in real-world multimode soft sensing environments.

A. Mode-Adaptive Multihead Attention

MHA enables simultaneous focus on various portions of a sequence, thereby modeling dependencies and preserving contextual continuity. This property makes MHA effective in identifying and representing dynamic features. However, conventional MHA is limited in its ability to model intermode dependencies, making it insufficient for capturing collaborative relationships across modes in complex industrial process data. To address this limitation, an MAMHA mechanism is proposed, which integrates a GMM and an adaptive weighting strategy within the MHA structure. By introducing distributional constraints and mode-aware adaptation, MAMHA

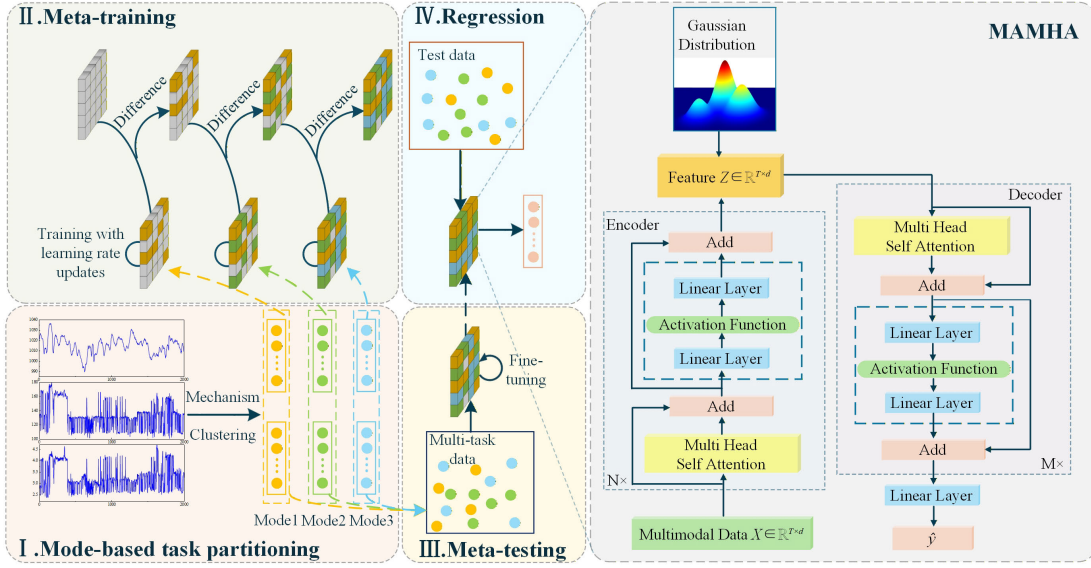


Fig. 2. Overall architecture of M-MAMHA.

enhances the ability to extract mode-specific information and model intermode relationships effectively.

The basic structure is defined as follows: 1) GMM for input classification and distribution constraints on features; 2) encoder for feature extraction; and 3) decoder for deep feature integration and regression. Assuming there is a multimode industrial process dataset, the dataset is first clustered based on operating conditions using the GMM to construct K mode subdatasets. At the same time, a representation of the Gaussian mixture distribution can be obtained

$$p(x) = \sum_{k=1}^K \pi_k \phi(x|\mu_k, \Sigma_k). \quad (9)$$

For each subdataset, the model inputs are formulated in the following way:

$$\begin{aligned} X(t) &= [\mathbf{x}(t), \mathbf{x}(t-1), \mathbf{x}(t-2), \dots, \mathbf{x}(t-T)] \\ &= \begin{bmatrix} x_1(t) & x_1(t-1) & x_1(t-2) & \dots & x_1(t-T) \\ x_2(t) & x_2(t-1) & x_2(t-2) & \dots & x_2(t-T) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_n(t) & x_n(t-1) & x_n(t-2) & \dots & x_n(t-T) \end{bmatrix} \end{aligned} \quad (10)$$

where T represents the length of each time window, while n corresponds to the number of input variable dimensions.

After feature extraction through the encoder, the hidden features of the data can be formulated as follows:

$$\mathbf{Z}(t) = \text{encoder}(\mathbf{X}(t)). \quad (11)$$

To encourage the encoder to effectively learn mode-specific features in the data, a KL divergence is introduced into the training loss function. Therefore, the loss function can be expressed as follows:

$$\begin{aligned} L &= L_{pre} + L_{KL} \\ &= \frac{1}{N} \sum_{i=1}^N \|y_i - y'_i\|^2 \end{aligned}$$

$$+ \frac{1}{N} \sum_{i=1}^N D_{KL}(q_k(\mathbf{Z} | \mathbf{X}_i) \| p_k(\mathbf{Z}_i)) \quad (12)$$

where y_i and y'_i represent real value and model output value, respectively. $q_k(\mathbf{Z} | \mathbf{X}_i)$ is the posterior distribution of feature output by the encoder and $p_k(\mathbf{Z}_i)$ is the corresponding mode-specific prior distribution in the latent space, which is modeled by a GMM. N refers to the total sample size within the dataset.

When training the model with data from each mode, a learnable adaptive weight ω_k is introduced for the KL divergence term of each mode. This weight indicates the significance of regularizing the feature distribution for each mode and allows the model to dynamically adjust its learning process. For the k th mode, the adaptive weighted KL divergence can be expressed as follows:

$$D_{KL}(q(\mathbf{Z} | \mathbf{X}) \| p(\mathbf{Z})) = \omega_k q_k(\mathbf{Z} | \mathbf{X}) \log \left(\frac{q_k(\mathbf{Z} | \mathbf{X})}{p_k(\mathbf{Z})} \right). \quad (13)$$

To implement the learning update for the weight ω_k , the gradient of the loss with respect to ω_k needs to be computed. Clearly, the loss gradient relative to ω_k is simply the unweighted KL divergence. Therefore, the gradient can be expressed as follows:

$$\frac{\partial D_{KL}}{\partial \omega_k} = q_k(\mathbf{Z} | \mathbf{X}) \log \left(\frac{q_k(\mathbf{Z} | \mathbf{X})}{p_k(\mathbf{Z})} \right). \quad (14)$$

Accordingly, the update formula for the adaptive weight can be written as follows:

$$\omega_k = \omega_k - \gamma \frac{\partial D_{KL}}{\partial \omega_k} \quad (15)$$

where γ represents the learning rate.

This adaptive KL divergence mechanism is employed exclusively during the meta-training phase, where the model learns from mode-specific tasks. This design encourages the encoder to effectively capture the statistical properties inherent to each mode, thereby enhancing feature alignment under

Algorithm 1 Meta for the Multimodal Soft Sensor**Input:**

Training task set $T = \{T_1, T_2, \dots, T_k\}$
 Testing task set $T_{\text{fine-tune}}$
 Initial model parameters ϕ
 Initial parameter-specific learning rate $\eta_i = [\eta_{i,1}, \eta_{i,2}, \dots, \eta_{i,d}]^\top$ in task T_i
 Meta-training learning rate γ_1 and γ_2
 Meta-testing learning rate η_{test}
 Number of meta-training iterations M
 Number of inner-loop iterations E
 Number of meta-testing iterations N

Output:

Updated model parameters ϕ
 Updated parameter-specific learning rate η_i
Meta-training stage:
for $m = 1, 2, \dots, M$ **do**
 for each task $T_i \in T$ **do**
 $\phi_i = \phi$
 for $e = 1, 2, \dots, E$ **do**
 Sample a data batch D_i^{train} from the data distribution of task T_i .
 Calculate the loss $L_{T_i}(\phi)$ on D_i^{train} .
 Compute the gradient of the loss $\nabla_{\phi} L_{T_i}(\phi)$.
 Update the task-specific parameters: $\phi'_i = \phi - \eta_i \odot \nabla_{\phi} L_{T_i}(\phi)$.
 end for
 Update the global parameters $\phi = \phi + \gamma_1 \cdot (\phi'_i - \phi)$.
 Update the parameter-specific learning rate $\eta_{i,j} = \eta_{i,j} - \gamma_2 \cdot \nabla_{\eta_{i,j}} L_{T_i}(\phi'_i)$.
 end for
end for
Meta-testing stage:
for $n = 1, 2, \dots, N$ **do**
 Further update the model parameters on test task set
 $\phi = \phi - \eta_{\text{test}} \cdot \nabla_{\phi} L_{T_{\text{fine-tune}}}(\phi)$.
end for

distributional constraints. Furthermore, by integrating a meta-learning framework, the model is optimized to rapidly adapt to new mode-specific data distributions. During the meta-testing phase, the model is fine-tuned on a mixture of modes without applying additional KL divergence regularization. This ensures that the learned representations can generalize across diverse mode data, improving cross-mode adaptability and predictive robustness.

B. Meta for Multimode Soft Sensor

The Meta4MSS framework, based on Reptile, is designed to optimize model performance in multimode soft sensing tasks. The framework consists of two key phases: meta-training and meta-testing. During the meta-training phase, each mode subdataset is treated as an independent meta-task and input into the model separately. Given the differences between tasks, each meta-training task is assigned a specific and adjustable learning rate for each parameter. This approach enables the model to adaptively adjust its learning strategy

by taking into account the specific distribution traits of each mode, allowing it to learn features that are tailored to each task. Additionally, the introduction of learnable parameters for each learning rate provides the flexibility to dynamically adjust the optimization process, improving the potential of the model in capturing complex relationships between the different modes. After completing meta-training, the model enters the meta-testing phase, where it is further trained using a dataset that includes data from all modes. This cross-mode training approach allows the model to capture the interdependencies and correlations across different modes, thereby enhancing its overall performance. The algorithm workflow of the Meta4MSS framework is shown as follows. The detailed implementation of the Meta4MSS framework is presented in Algorithm 1.

To update the parameter-specific learning rate, it is necessary to calculate the gradient with respect to the learning rate. The gradient update rule is as follows:

$$\phi'_i = \phi - \eta_i \odot \nabla_{\phi} L_{T_i}(\phi). \quad (16)$$

For the meta-loss $L_{T_i}(\phi'_i)$, the objective is to find the learning rate η_i that minimizes the meta-loss. Therefore, it is necessary to compute

$$\nabla_{\eta_{i,j}} L_{T_i}(\phi'_i) = \frac{\partial L_{T_i}(\phi'_i)}{\partial \eta_{i,j}}. \quad (17)$$

Applying the chain rule, it can be obtained

$$\frac{\partial L_{T_i}(\phi'_i)}{\partial \eta_{i,j}} = \frac{\partial L_{T_i}(\phi'_i)}{\partial \phi'_i} \cdot \frac{\partial \phi'_i}{\partial \eta_{i,j}}. \quad (18)$$

Given $\phi'_i = \phi - \eta_i \odot \nabla_{\phi} L_{T_i}(\phi)$

$$\frac{\partial \phi'_i}{\partial \eta_{i,j}} = -\nabla_{\phi} L_{T_i}(\phi). \quad (19)$$

Therefore, there is

$$\nabla_{\eta_{i,j}} L_{T_i}(\phi'_i) = -\frac{\partial L_{T_i}(\phi'_i)}{\partial \phi'} \cdot \nabla_{\phi} L_{T_i}(\phi). \quad (20)$$

Finally, the update rule for the learning rate is

$$\eta_{i,j} = \eta_{i,j} - \gamma_2 \cdot \nabla_{\eta_{i,j}} L_{T_i}(\phi'_i) \quad (21)$$

where γ_2 denotes the meta-learning rate used to update the parameter-specific learning rate η_i .

C. Deployment of M-MAMHA

To efficiently model multimode data in industrial processes, this article proposes an advanced model, M-MAMHA. The application of M-MAMHA in soft sensing modeling follows these steps.

Step 1: Obtain the industrial process dataset $D = \{X, Y\}$. To eliminate the impact of different data scales, standardization is employed for the data. Next, the dataset is split into training and testing sets. Based on the actual operating conditions of the dataset or using clustering algorithms, the training set is further divided into k subsets, corresponding to k mode subsets. A certain number of samples are extracted from each subset to construct k meta-tasks $T = \{T_1, T_2, \dots, T_k\}$ for the meta-training phase of the Meta4MSS framework. Additionally, another set of samples is extracted from each

subset to form the overall task $T_{\text{fine-tune}}$ for the meta-testing phase of the Meta4MSS framework.

Step 2: Initialize the model parameters and learning rates required for meta-learning within the Meta4MSS framework.

Step 3: Within the meta-training phase, the model is trained on meta-tasks in T , learning and adjusting parameters to identify complex relationships and mode information. By training across multiple meta-tasks, the model can capture potential features and mode variations in the data, thereby enhancing the performance of the soft sensing model.

Step 4: In the meta-testing phase, the model that has undergone meta-training is further trained on the overall task $T_{\text{fine-tune}}$. This phase of training utilizes the meta-tasks $T_{\text{fine-tune}}$ to perform deeper learning iterations, optimizing model parameters so that the model better understands the potential relationships and mode information within the multimode data.

Step 5: Evaluate the performance of the model to ensure its effectiveness in practical applications.

Three performance metrics are employed to evaluate the model's performance in multimode soft sensing applications: RMSE, MAE, and R^2 . The calculation formulas are as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y'_i - y_i)^2} \quad (22)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y'_i - y_i| \quad (23)$$

$$R^2 = \frac{\sum_{i=1}^n (y'_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n (y_i - y'_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (24)$$

where n corresponds to the sample size, y' to the predicted output, and $\bar{y}$ to the average of the ground-truth values.

IV. CASE STUDY

In this section, two real industrial process datasets, SRU [3] and CCP [8], are utilized to validate the efficiency of the M-MAMHA. Five benchmark models are established for comparison, including two traditional models (GRU and MHA), as well as three advanced models (DPTR [22], ICF-SAE [23], and DMVAER [25]), to highlight the superiority of the proposed method. Notably, the two traditional models treat process data as a single mode, while the three advanced models consider process data as multimode. In addition, ablation experiments are conducted to systematically analyze the impact of the proposed MAMHA and Meta4MSS framework on model performance.

To ensure the reliability of the experimental evaluation, all models are independently trained and tested ten times using different random seeds. The reported RMSE, MAE, and R^2 values represent the mean results across these ten repeated runs. The hardware configuration includes a 13th Gen Intel¹ Core² i5-13600KF CPU running at 3.50 GHz, an NVIDIA GeForce RTX 3060 Ti GPU, 16 GB of DDR4 memory (2 × 8 GB), and dual 1 TB SSDs. The computing environment consists of a 64-bit Windows 11 operating

system, with PyTorch 2.2.2 and CUDA 12.6 as the deep learning framework, and Python version 3.10.13.

A. SRU Process

Within the industrial petroleum refining process, certain acid gases that pollute the air and harm human health are generated. To reduce the emission of these pollutants, the SRU is designed to eliminate contaminants in the acid gases, while the recovered elemental sulfur from the reactions is used as a valuable by-product. The SRU reactor consists of two combustion chambers, each processing a different type of acid gas: one is MEA gas, rich in H_2S , and the other is SWS gas, rich in both H_2S and NH_3 . To ensure the reactions proceed efficiently, the concentrations of H_2S and SO_2 in the tail gas must be continuously monitored in real-time. However, due to the corrosive nature of these acid gases, hardware sensors often demand frequent removal and maintenance. This not only disrupts real-time control of the reaction but also increases production costs. Therefore, designing a soft sensing model capable of predicting the concentrations of H_2S and SO_2 in the tail gas is highly necessary. This classic soft sensing dataset, which comprises time-series data collected over extended periods, provides a valuable basis for developing predictive models.

In the experiment, five relevant process variables from the dataset were selected as input variables, and SO_2 was chosen as the output variable. The dataset contains a total of 100 071 samples, of which 70% were used for training, while the remaining samples were reserved for testing. Considering the dynamic characteristics of the process, the variables were processed as follows:

$$y(t) = f(x_i(t), x_i(t-5), x_i(t-7), x_i(t-9)) \quad (25)$$

where $y(t)$ represents the SO_2 concentration at time t and $x_i(t)$, $i = 1, 2, 3, 4, 5$, denotes the corresponding process variables. Additionally, $x_i(t-5)$, $x_i(t-7)$, and $x_i(t-9)$ represent the lagged values of these variables with sampling intervals of 5, 7, and 9, respectively. As a result, the input data dimension is expanded from 5 to 20. In addition, based on the actual operating conditions, the training data is divided into five mode subdatasets using the GMM.

To ensure the fairness and validity of the experimental setup, the hyperparameters of each model were optimized using a grid search algorithm. The main hyperparameter configurations for each model are as follows: GRU has 12 hidden nodes. MHA utilizes 4 heads, with 2 hidden nodes per head. For DPVR, the structures of EncoderNet and PriorNet are [26, 46], while AdversaryNet and DecoderNet are configured as [12, 8]. The number of neurons in the latent space is 31. And the SAE structure of ICF-SAE is [20-16-13-16-20]. For DMAVER, the dimensions of global and local latent variables are set as [16, 8], and both the GRU hidden size and the number of hidden neurons in the decoder are chosen as 16. Both the encoder and decoder of M-MAMHA consist of 4 layers, with 5 heads, each head containing 4 hidden nodes. The feedforward layer has 44 hidden nodes.

The simulation results of all models are summarized in Table I, while Fig. 3 presents the distribution plots of predicted versus true values, offering a visual assessment of

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²Trademarked.

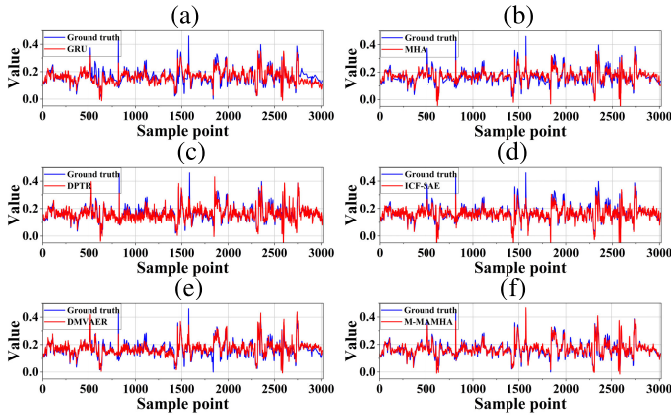


Fig. 3. Distribution plots of all models on the SRU. (a) GRU. (b) MHA. (c) DPTR. (d) ICF-SAE. (e) DMVAER. (f) M-MAMHA.

TABLE I
SIMULATION RESULT OF THE SRU

METHOD	RMSE	MAE	R^2
GRU	0.0313	0.0243	0.5709
MHA	0.0281	0.0217	0.6099
DPTR [22]	0.0285	0.0216	0.6612
ICF-SAE [23]	0.0263	0.0196	0.6902
DMVAER [25]	0.0262	0.0205	0.7197
M-MAMHA	0.0200	0.0155	0.8359

the predictive performance of each model. As shown in Table I, the proposed M-MAMHA model achieves the best performance across all three evaluation metrics. This superior performance highlights the effectiveness of partitioning process data by mode during training, which enables more specialized feature extraction and integration. By treating each mode as a distinct learning task, M-MAMHA can capture mode-specific information while simultaneously modeling intermode dependencies, leading to more accurate and robust predictions. In contrast, traditional models such as the GRU and MHA demonstrate a basic ability to follow key quality trends, as seen in Fig. 3, but their simplistic architectures, which treat the entire dataset as a homogeneous mode, limit their capacity to extract and exploit multimode information. DPTR incorporates transfer learning to alleviate distribution shifts between modes and successfully captures some mode features. However, its performance heavily depends on the availability of labeled data for each mode, and its relatively shallow architecture constrains its ability to generalize across complex multimode settings. ICF-SAE improves robustness to noise and multimode through SAEs, yet its reliance on deep unsupervised feature extraction—without explicit mode-aware training—hinders its ability to distinguish between different operational modes. DMVAER introduces a dynamic probabilistic framework that combines global mixture-based latent variables with local temporal representations, allowing it to model both mode distributions and dynamic behaviors. Despite this, it does not explicitly partition training data by mode, which reduces its effectiveness in learning mode-specific representations. Moreover, its generative modeling objective emphasizes input reconstruction over predictive

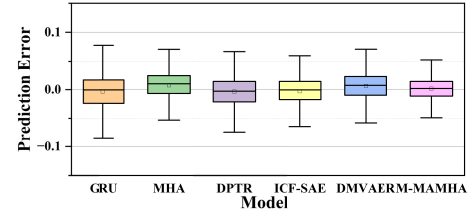


Fig. 4. Prediction error boxplots of all models on the SRU.

accuracy, which can lead to suboptimal results in supervised regression tasks.

To further evaluate the reliability of the performance improvements achieved by the proposed M-MAMHA method, boxplots of prediction errors are constructed based on the test set (Fig. 4), followed by a statistical significance analysis. In addition to visually demonstrating that M-MAMHA yields a markedly lower median error and a narrower interquartile range compared with all benchmark models, paired t-tests were conducted between M-MAMHA and each benchmark model. The resulting p-values were consistently below 0.001, indicating that the performance gains brought by M-MAMHA are statistically significant rather than attributable to random variability. These findings confirm that the proposed approach not only improves overall predictive accuracy but also exhibits more stable behavior across most samples, demonstrating strong robustness and reliability.

To further evaluate the practicality of the proposed framework, the average computational cost during meta-training, meta-testing, and inference is reported. On the SRU dataset, the model requires only 0.01 s per task per epoch during meta-training, 0.20 s per epoch during meta-testing, and 0.06 s for a single forward inference. It is worth noting that, thanks to the adopted task-wise pretraining followed by a light global fine-tuning strategy, the model exhibits fast convergence during meta-testing and typically requires only a very small number of epochs to reach optimal performance. This greatly reduces overall training time even in multitasking settings. These results indicate that the proposed method not only achieves strong predictive accuracy and generalization capability but also maintains high computational efficiency. This combination makes the method highly suitable for deployment in real-time industrial soft-sensing systems that demand rapid model updates and fast, high-precision predictions.

B. Combined Cycle Power Plant

The CCPP leverages the fast start-up, shutdown, and load adjustment of gas turbines to drive steam turbines for efficient power generation. On the other hand, the combustion process also leads to significant emissions of nitrogen oxides (NO_x) and carbon monoxide (CO), which contribute to global environmental issues such as photochemical smog, acid rain, and global warming. In response, the European Union has established strict regulations to control NO_x and CO emissions, mandating levels be kept below 25 ppmv. Accurate monitoring of these emissions in industrial processes is essential to ensure compliance with these regulations. Using soft sensing technology to model the relationship between other easily

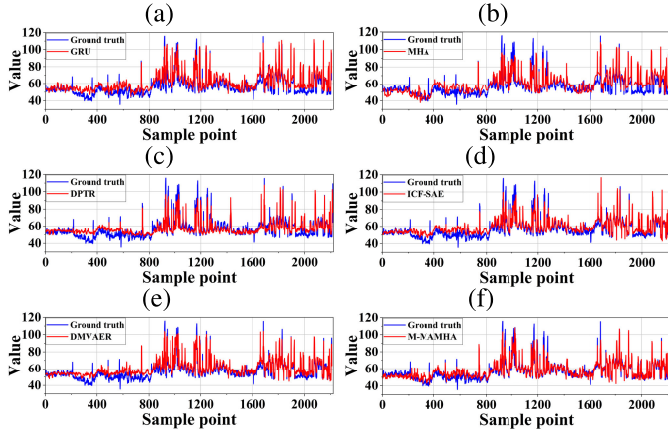


Fig. 5. Distribution plots of all models on CCPP. (a) GRU. (b) MHA. (c) DPTR (d) ICF-SAE. (e) DMVAER. (f) M-MAMHA.

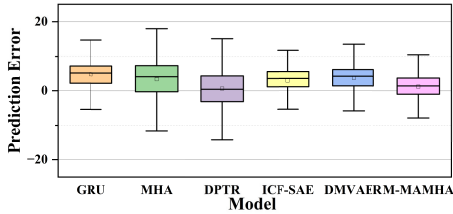


Fig. 6. Prediction error boxplots of all models on CCPP.

measurable variables and the NO_x and CO emissions offers a feasible and effective approach to continuous monitoring and emissions control.

This industrial emissions dataset consists of nine input variables and two key quality variables. The dataset contains hourly averaged sensor readings from 36 733 samples collected over five years, covering three operational load conditions ranging from partial (75%) to full load (100%). In the experiment, NO_x was chosen as the target variable, and 7384 samples collected in 2015 were used for model training, with 70% of the samples allocated for training and 30% for testing. In addition, based on the operational load conditions, the training data is divided into three mode subdatasets using the GMM. Similarly, grid search optimization was employed to tune the hyperparameters of each model. Among them, GRU has 15 hidden nodes. MHA utilizes 3 heads, with 4 hidden nodes per head. For DPVR, the structures of EncoderNet and PriorNet are [32, 64], while AdversaryNet and DecoderNet are configured as [14, 7]. The number of neurons in the latent space is 64. And the SAE structure of ICF-SAE is [9-14-20-14-9]. For DMAVER, the dimensions of global and local latent variables are set as [10,5], and both the GRU hidden size and the number of hidden neurons in the decoder are chosen as 12. Both the encoder and the decoder of M-MAMHA consist of 4 layers, with 4 heads, each head containing 5 hidden nodes. The feedforward layer has 22 hidden nodes.

The simulation results and distribution plots are presented in Table II and Fig. 5, respectively. As shown in Table II, the proposed M-MAMHA consistently outperforms all baseline models on the CCPP dataset, achieving superior performance across all three evaluation metrics. Traditional models, which treat the process data as a single mode, fail to leverage the underlying multimode information, resulting in relatively poor

TABLE II
SIMULATION RESULT OF CCPP

METHOD	RMSE	MAE	R^2
GRU	6.7369	5.5551	0.6107
MHA	7.1628	5.5710	0.5754
DPTR [22]	5.5406	4.3499	0.5462
ICF-SAE [23]	5.3329	4.3329	0.6181
DMVAER [25]	5.5427	4.6220	0.6853
M-MAMHA	4.7083	3.3972	0.7544

prediction accuracy. This highlights the effectiveness and feasibility of mode-based data partitioning during model training. While ICF-SAE and DPTR consider the multimode nature of process data and achieve noticeable improvements in the RMSE and MAE, they process each sampling point independently, limiting the ability to capture temporal dependencies and dynamic behavior—reflected in the lower R^2 scores. DMVAER incorporates a dynamic variational inference framework and achieves comparatively strong performance in terms of the R^2 metric, demonstrating its ability to model dynamic characteristics. However, its failure to explicitly partition the training data by mode leads to weaker performance in error-based metrics. In Fig. 6, the proposed M-MAMHA model demonstrates the most favorable combination of performance indicators, which indicates that M-MAMHA produces not only more accurate predictions on average, but also maintains a high level of consistency across different samples.

To further substantiate the performance gains of M-MAMHA, boxplots of prediction errors Fig. 6 are analyzed together with statistical significance testing. As shown in Fig. 6, M-MAMHA exhibits a clearly lower median error and a notably narrower IQR than all baselines. Paired t-tests between M-MAMHA and each benchmark model yielded p-values below 0.01, indicating that the improvements are statistically significant rather than due to random variation. These results confirm that M-MAMHA is both more accurate and more stable in prediction tasks.

The proposed model framework and training strategy also demonstrate high training and inference efficiency on the SRU dataset. The meta-training, meta-testing, and inference times are 0.03, 0.19, and 0.04 s, respectively. The results further demonstrate that the proposed method delivers accurate predictions with minimal computation overhead, highlighting its suitability for real-time industrial soft-sensing applications.

C. Ablation Experiment

To investigate the effect of MAMHA and the Meta4MSS framework, ablation experiments were carefully designed. For the ablation study, the proposed M-MAMHA model is used as the baseline, and comparison models are constructed by sequentially removing key modules. Specifically, the first comparison model (w/o MAMHA) removes the adaptive KL divergence while retaining the M4MSS module. The second model (w/o M4MSS) replaces the M4MSS module with the traditional Reptile algorithm but retains the adaptive KL divergence mechanism. Finally, the third model (w/o MAMHA and M4MSS) removes both the adaptive KL divergence and

TABLE III
SIMULATION RESULT OF ABLATION EXPERIMENTS

METHOD	SRU			CCPP		
	RMSE	MAE	R ²	RMSE	MAE	R ²
Baseline	0.0200	0.0155	0.8359	4.7083	3.3972	0.7544
w/o M4MSS	0.0222	0.0165	0.7894	5.0118	3.5314	0.6987
w/o MAMHA	0.0245	0.0184	0.7464	5.4919	3.8050	0.6538
w/o MAMHA & M4MSS	0.0271	0.0208	0.6963	5.6362	4.0723	0.6216

the M4MSS module, instead relying solely on the traditional Reptile algorithm. All three ablation models use the same hyperparameter settings as M-MAMHA and are simulated on both the SRU and CCPP datasets for effective comparison.

The simulation results are presented in Table III. It is evident that the two modules in M-MAMHA have a significant impact on the overall performance of the model. Taking the full model as the baseline, removing Meta4MSS (w/o M4MSS) results in a performance degradation of 11.0% (RMSE), 6.5% (MAE), and 5.6% (R²) on the SRU dataset and 6.4% (RMSE), 3.9% (MAE), and 7.4% (R²) on the CCPP dataset. This indicates that the parameter-specific learning rate in Meta4MSS substantially improves training adaptability across modes. Removing MAMHA (w/o MAMHA) yields an even larger performance drop: 22.5% (RMSE), 18.7% (MAE), and 10.7% (R²) on SRU and 16.7% (RMSE), 12.0% (MAE), and 13.4% (R²) on CCPP. These results confirm that the adaptive KL divergence plays a crucial role in aligning mode-specific feature distributions and enhancing predictive stability in multimode environments. The combined removal of both components (w/o MAMHA and M4MSS) leads to the most severe degradation, with performance reductions reaching 35.5% (RMSE), 34.2% (MAE), and 16.7% (R²) on SRU and 19.7% (RMSE), 19.9% (MAE), and 17.6% (R²) on CCPP. These results reflect that MAMHA and Meta4MSS provide complementary benefits: MAMHA strengthens intramode representation learning through probabilistic alignment, while Meta4MSS enhances cross-mode adaptability via task-specific optimization dynamics. Overall, the quantitative analysis clearly demonstrates that the two modules together enable the full M-MAMHA model to achieve the highest accuracy and robustness across both datasets.

V. CONCLUSION

In this article, a novel M-MAMHA method is proposed to address the limited capacity of capturing mode-specific information and modeling the dynamic dependencies among process variables in industrial applications. First, a MAMHA model is introduced, which utilizes an encoder-decoder structure based on MHA for feature extraction. At the same time, the model incorporates a Gaussian distribution into the feature space to better characterize mode-specific statistical properties within the data. To further enhance feature alignment, a learnable adaptive weight is introduced for the KL divergence term, enabling the model to dynamically regulate the strength of distribution regularization during training. Then, a Meta4MSS framework is developed, which treats each mode as a separate meta-task to train the MAMHA. A parameter-specific learning rate is introduced to training stage to allow the model to

automatically adjust the learning rate according to mode heterogeneity, thereby improving convergence efficiency and promoting cross-mode generalization. Finally, the proposed method is applied to two multimode industrial process data soft sensing modeling tasks, demonstrating the effectiveness of M-MAMHA.

Despite the promising results, the current work still faces several limitations. First, the imbalance of sample sizes across different modes may influence model stability and limit performance under extreme distribution shifts. Second, although the computational cost is acceptable for practical deployment, further optimization is desirable for large-scale or real-time applications. In future work, few-shot and transfer learning techniques will be explored to alleviate mode imbalance and improve data efficiency, and more lightweight strategies will be investigated to enhance runtime performance. These directions are expected to broaden the applicability of the proposed method to a wider range of multimode industrial scenarios.

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